

Systems Programming 1 – COMP2019

Tutorial and Lab Practice Two

This work **will be marked** and is worth **4%** of your overall assessment. It is due by **Sunday, 22nd of February**. and Please submit your work online via vUWS. You should explain your answers and demonstrate your programs to your Tutor during your next scheduled lab session after the due date. **Email submission is not accepted.**

Tutorial

1. Get any book/resource about C programming language, read the chapters regarding C data types and structures, pointers and arrays, stacks.
2. Review the terminology introduced and concepts taught in lecture 2.
3. Define an enumerated data type which contains all the week days.
4. Using the following **struct**:

```
typedef char DATA;
struct node
{
    DATA d;
    struct node *next;
};
```

Write pseudo code for building a linked list using a loop. Your list should be built in such a way that the new node is added at the beginning of the list. **(1%)**

Lab Practice

1. Using the following **struct**:

```
typedef char DATA;
struct node
{
    DATA d;
    struct node *next;
};
```

Write a program to build a single linked list of 10 nodes using a loop. You may assign any character to the data field (d). Your list should be built in such a way that the new node is added at the end of the list. **(1%)**

2. Add a function to your program to print the list. **(1%)**
3. Write a function to reverse the order of the list. Print the list in both its original and reversed orders. **(1%)**
4. Revise your program so that the generated list has the opposite order, i.e. the new node is added at the beginning of the list.

Optional Work (For those students who wish to practise more.)

1. Write a program to build a double linked list using a loop. The struct should look like:

```
typedef char DATA;  
struct node  
{  
    DATA d;  
    struct node *next;  
    struct node *prev;  
};
```

2. Write a C program to print out the pyramid of stars. Ask users to enter the number of rows to be printed at run time. This is an example of 3 rows.

```
*  
**  
***
```